

All Six Open Questions on the Differentiation Operator

A norm counterexample and universal power, range, and compactness obstructions

Automated mathematical discovery run `fa_banach_001`
Agent `agent_lane_14` Model GPT5.6

22 July 2026

Status: full solution of all six questions, likely valid, pending human review.
The packet answers every question in Section 6 of Allen–Jackson [1], including exact spectra on the weighted Banach and discrete Hardy spaces.

1 Result at a glance

Let T be the locally finite rooted tree of the source paper, with every vertex having a child, let $b(o) = o$ and $b(v) = v^-$ for $v \neq o$, and put

$$C_b f = f \circ b, \quad Df = f - f \circ b.$$

On a discrete function space, point evaluations are continuous and the functions separate every pair of vertices.

Question	Answer	Mechanism
1	Yes	A weighted Hilbert space has $\ C_b\ = \sqrt{2}$ but $\ D\ = 2$.
2	No	Pointwise bounded D -orbits contain only constant functions; hence D is never power-bounded.
3	No	Every function in $\text{ran}(C_b)$ agrees at the root and each root child.
4	No	A nonzero spectral point of D is not an eigenvalue, ruling out compactness.
5	Exact formulas	A growth-radius disk plus the root point for weighted spaces; the unit disk for Hardy spaces.
6	No	The compactness obstruction is universal, hence applies to the intended path-tree spaces.

Thus the norm conjecture in the source is false, while its isometry conjecture is true in the stronger power-bounded form.

2 Source questions

The questions occur in Section 6, page 17 of arXiv:2207.10211 [1]. The following is a full-width crop of the source page.

The relevant questions ask: whether $\|D\|$ can differ from $1 + \|C_b\|$; whether D can act isometrically; whether C_b can be surjective; whether D can be compact; and whether D is compact on the Lipschitz or weighted Banach space of a path tree.

6. OPEN QUESTIONS

We conclude with open questions which were inspired while developing this manuscript. Our hope is that these questions will initiate further research in this area of operator theory.

1. Is there a discrete function space on which D is bounded, but its norm is not $1 + \|C_b\|$?
2. Is there a non-trivial discrete function space on which D acts isometrically?
3. Is there a non-trivial discrete function space on which C_b is surjective?
4. Is there a non-trivial discrete function space on which D is compact?
5. What is the spectrum of D or C_b acting on either L_μ^∞ or $\mathbb{T}_p(T)$?
6. Is D compact on $\mathcal{L}(T)$ or $L_\mu^\infty(T)$ for path tree T ?

ACKNOWLEDGEMENTS

Figure 1: The six open questions in the source paper. This packet answers all six.

3 Definitions and conventions

A *discrete function space* $\mathcal{X}$ on T is a complex Banach space of complex-valued functions on T such that:

1. every evaluation $\delta_v(f) = f(v)$ is continuous;
2. if $f(v) = f(w)$ for every $f \in \mathcal{X}$, then $v = w$.

The second condition is the point-separation axiom. The source assumes that every vertex has a child. This child condition, rather than homogeneity or a specific branching number, is exactly what the proofs below use.

Whenever D maps $\mathcal{X}$ into itself, the source's closed graph argument makes D bounded, and $C_b = I - D$ is bounded as well [1, Theorems 2.6–2.7].

4 Proof intuition

Four elementary geometric features drive all six answers.

First, distribute one unit of Hilbert mass from every parent among its children. After conjugating the resulting weighted ℓ_2 space to ordinary $\ell_2(T)$, the operator C_b becomes $P_o + U$, where P_o is the root projection and U shifts mass one level outward isometrically. The extra root copy raises $\|C_b\|$ to $\sqrt{2}$, while alternating long U -orbits show that $\|I - P_o - U\| = 2$. This breaks the conjectured norm identity.

Second, repeated differences reveal any nonzero edge increment one generation later as a term growing linearly with the iteration number. Thus a function with a pointwise bounded D -orbit must be constant. Norm power-boundedness would make every orbit pointwise bounded, which is incompatible with point separation.

Third, the root and each root child lie in the same fibre of b . Hence every function in the range of C_b identifies those two vertices, so C_b cannot be onto a point-separating space. This range obstruction puts 1 in the spectrum of D but not in its point spectrum, which compact operator theory forbids.

Fourth, both named spaces split as ℓ_∞ -sums of their finite dimensional level spaces. On the nonroot tail, C_b is an operator-weighted unilateral shift. A one-level resolvent test shows that the spectrum of this shift is exactly the disk determined by its uniform asymptotic block-growth rate.

For L_μ^∞ this rate is a descendant/ancestor weight ratio; for $\mathbb{T}_p$ every level map is isometric, so the rate is one.

5 A Hilbert-space counterexample to the norm identity

Theorem 1 (Answer to Question 1). *On every rooted locally finite tree in which each vertex has a child, there is a discrete Hilbert function space H_w such that*

$$\|C_b\| = \sqrt{2}, \quad \|D\| = 2.$$

Consequently $\|D\| \neq 1 + \|C_b\|$.

Proof. For every nonroot vertex v , choose $p_v > 0$ so that

$$\sum_{v \in \text{ch}(u)} p_v = 1 \quad (u \in T).$$

This is possible because every child set is finite and nonempty. Define $w(o) = 1$ and recursively

$$w(v) = p_v w(v^-), \quad v \neq o.$$

All weights are positive. Let

$$H_w = \left\{ f : T \longrightarrow \mathbb{C} : \|f\|_w^2 := \sum_{v \in T} w(v) |f(v)|^2 < \infty \right\}.$$

This is a Hilbert space. Evaluation at v has norm $w(v)^{-1/2}$, and the finitely supported point masses belong to H_w and separate the vertices. Thus H_w is a discrete function space.

The map $J : H_w \rightarrow \ell_2(T)$ given by

$$(Jf)(v) = w(v)^{1/2} f(v)$$

is unitary. Define U on $\ell_2(T)$ by

$$(Uy)(o) = 0, \quad (Uy)(v) = p_v^{1/2} y(v^-) \quad (v \neq o).$$

Then

$$\begin{aligned} \|Uy\|_2^2 &= \sum_{u \in T} \sum_{v \in \text{ch}(u)} p_v |y(u)|^2 \\ &= \sum_{u \in T} |y(u)|^2, \end{aligned}$$

so U is an isometry. Let P_o be the orthogonal projection onto the root coordinate. Directly from the definitions,

$$JC_b J^{-1} = P_o + U, \quad JDJ^{-1} = I - P_o - U = (I - P_o)(I - U).$$

The vectors $P_o y$ and Uy have disjoint supports. Therefore

$$\|(P_o + U)y\|_2^2 = \|P_o y\|_2^2 + \|y\|_2^2 \leq 2\|y\|_2^2.$$

Equality holds for the root unit vector e_o , so $\|C_b\| = \sqrt{2}$.

The displayed factorization gives $\|D\| \leq 2$. To obtain the reverse inequality, observe that $U^k e_o$ is a unit vector supported on level k ; hence these vectors are orthonormal. For $N \geq 1$, put

$$y_N = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (-1)^k U^k e_o.$$

Then $\|y_N\|_2 = 1$. After applying $(I - P_o)(I - U)$, the coefficients on levels $1, \dots, N-1$ have modulus $2/\sqrt{N}$, and the final coefficient on level N has modulus $1/\sqrt{N}$. Hence

$$\|Dy_N\|_2^2 = \frac{4(N-1) + 1}{N} = 4 - \frac{3}{N}.$$

Letting $N \rightarrow \infty$ yields $\|D\| \geq 2$. Thus $\|D\| = 2$, whereas $1 + \|C_b\| = 1 + \sqrt{2}$. $\square$

6 Iterated differences cannot be pointwise bounded

Lemma 2 (Pointwise growth lemma). *Let $f : T \rightarrow \mathbb{C}$ be any function. If, for every $v \in T$, the scalar sequence $\{(D^n f)(v) : n \geq 0\}$ is bounded, then f is constant.*

Proof. We prove by induction on $m \geq 1$ that $f(v) = f(v^-)$ for every vertex v of depth m . Fix such a v and choose a child z of v . Along the path from the root through v to z , write

$$a_j = f(v_j) - f(v_{j-1}), \quad o = v_0, v_1, \dots, v_m = v, v_{m+1} = z.$$

For $m = 1$, $D^n f(o) = 0$ for every $n \geq 1$. For the induction step, the earlier edge increments vanish by hypothesis, which again gives $D^n f(v_{m-1}) = 0$ for all $n \geq 1$. In either case the recurrence

$$(D^{n+1} f)(x) = (D^n f)(x) - (D^n f)(x^-)$$

shows that

$$(D^n f)(v) = a_m \quad (n \geq 1).$$

Applying the same recurrence at z gives

$$(D^n f)(z) = a_{m+1} - (n-1)a_m \quad (n \geq 1).$$

The left side is bounded in n , so $a_m = 0$. Since v was arbitrary, the induction closes at every depth. All edge increments vanish, and connectedness makes f constant. $\square$

Theorem 3 (Answer to Question 2). *The differentiation operator is not power-bounded on any discrete function space. In particular, it is never an isometry.*

Proof. Suppose D were power-bounded on $\mathcal{X}$, say $\sup_n \|D^n\| \leq M$. For $f \in \mathcal{X}$ and $v \in T$, continuity of evaluation gives

$$|(D^n f)(v)| \leq \|\delta_v\| M \|f\|.$$

Lemma 2 makes every $f \in \mathcal{X}$ constant. Such a space cannot separate the root from a child, contradicting the definition of a discrete function space. An isometry has $\|D^n\| = 1$ for every n , so it is excluded. $\square$

7 The backward composition operator is never onto

Theorem 4 (Answer to Question 3). *The operator C_b is not surjective on any discrete function space.*

Proof. Let v be any child of the root. Every $g = C_b f$ satisfies

$$g(o) = f(b(o)) = f(o) = f(b(v)) = g(v).$$

If C_b were onto $\mathcal{X}$, every member of $\mathcal{X}$ would take equal values at the distinct vertices o and v . This contradicts point separation. $\square$

This sharpens the source paper's observation that surjectivity would force range functions to be constant on sibling sets: the fibre $b^{-1}(o)$ already contains both the root and every root child.

8 Differentiation is never compact

We first close a finite-dimensional loophole needed for the compactness argument.

Lemma 5. *There is no finite-dimensional discrete function space $\mathcal{X}$ for which $D\mathcal{X} \subseteq \mathcal{X}$.*

Proof. The source proves that the only possible eigenvalue of D is 0, and that the corresponding eigenvectors are constant functions [1, Theorem 2.14]. For completeness, a nonzero eigenvalue is also ruled out directly by evaluating $Df = \lambda f$ first at the root and then inductively along the tree.

If $\mathcal{X}$ were finite-dimensional and nonzero, complex finite-dimensional spectral theory would therefore make D nilpotent. Let m be its nilpotency index. If $m \geq 2$, choose f with $g = D^{m-1}f \neq 0$. Then $Dg = 0$, so g is constant. But g lies in the range of D , and every function in that range vanishes at the root. Thus $g = 0$, a contradiction. Hence $m = 1$ and $D = 0$. Then every member of $\mathcal{X}$ is constant, again contradicting point separation. $\square$

Theorem 6 (Answers to Questions 4 and 6). *The differentiation operator is not compact on any discrete function space. Consequently it is not compact on either the Lipschitz space or any weighted Banach space of the intended path tree for which it acts boundedly.*

Proof. Suppose D were compact on $\mathcal{X}$. Then D is bounded, so $C_b = I - D$ is bounded. Lemma 5 makes $\mathcal{X}$ infinite-dimensional.

By Theorem 4, C_b is not invertible. Therefore

$$0 \in \sigma(C_b), \quad 1 \in \sigma(D),$$

the latter because $D - I = -C_b$. On the other hand, C_b is injective: for each $u \in T$, choose a child v with $b(v) = u$; if $C_b f = 0$, then $f(u) = f(b(v)) = 0$. Hence 1 is not an eigenvalue of D .

Every nonzero spectral value of a compact operator on an infinite-dimensional complex Banach space is an eigenvalue [2, Chapter VII, Section 4]. The spectral point 1 contradicts this theorem. Thus D cannot be compact.

For a named path-tree space, an unbounded D is automatically not compact; when D is bounded, the universal conclusion just proved applies. $\square$

9 Exact spectra on the weighted and Hardy spaces

We begin with a self-contained shift lemma. It isolates the only spectral calculation needed for both families.

Lemma 7 (Operator-weighted unilateral shift). *Let $(E_n)_{n \geq 1}$ be nonzero complex Banach spaces and let $B_n : E_{n-1} \rightarrow E_n$ ($n \geq 2$) be uniformly bounded. On*

$$Z = \left(\bigoplus_{n \geq 1} E_n \right)_{\ell_\infty}$$

define

$$(Sx)_1 = 0, \quad (Sx)_n = B_n x_{n-1} \quad (n \geq 2).$$

For $k \geq 1$, put

$$R_k = \sup_{n \geq 1} \|B_{n+k} B_{n+k-1} \cdots B_{n+1}\|, \quad \rho = \lim_{k \rightarrow \infty} R_k^{1/k}.$$

Then the limit exists and

$$\sigma(S) = \{\lambda \in \mathbb{C} : |\lambda| \leq \rho\}.$$

Proof. The sequence (R_k) is submultiplicative, and direct calculation gives $\|S^k\| = R_k$. Thus the limit defining ρ exists and equals the spectral radius of S . In particular, the spectrum is contained in the stated disk.

The operator S is not onto because the first coordinate of every range vector vanishes, so $0 \in \sigma(S)$. Now fix $0 < |\lambda| < \rho$ and suppose, for a contradiction, that $\lambda I - S$ has a bounded inverse of norm C . Fix $n \geq 1$ and a unit vector $y_n \in E_n$, and let $y \in Z$ have this as its only nonzero coordinate. For $x = (\lambda I - S)^{-1}y$, the triangular recurrence first gives $x_j = 0$ for $j < n$, and then

$$x_{n+k} = \lambda^{-k-1} B_{n+k} B_{n+k-1} \cdots B_{n+1} y_n \quad (k \geq 0),$$

where the empty product for $k = 0$ is the identity. Since $\|x\| \leq C$,

$$\|B_{n+k} \cdots B_{n+1} y_n\| \leq C |\lambda|^{k+1}.$$

Taking the supremum over unit y_n and over n yields $R_k \leq C |\lambda|^{k+1}$, hence $\rho \leq |\lambda|$, a contradiction. Therefore the open disk of radius ρ lies in the spectrum. Closedness of the spectrum supplies its boundary. (When $\rho = 0$, the spectral-radius inclusion and $0 \in \sigma(S)$ give the same conclusion.) $\square$

9.1 Weighted Banach spaces

Let $\mu : T \rightarrow (0, \infty)$ be a weight and suppose

$$M_\mu := \sup_{v \in T^*} \frac{\mu(v)}{\mu(v^-)} < \infty,$$

which is exactly the boundedness condition for D (equivalently C_b) on $L_\mu^\infty(T)$ in the source. For $k \geq 1$, define

$$R_k(\mu) = \sup \left\{ \frac{\mu(v)}{\mu(u)} : u \in T^*, v \text{ is a descendant of } u, |v| - |u| = k \right\},$$

and

$$\rho_\mu = \lim_{k \rightarrow \infty} R_k(\mu)^{1/k}.$$

The limit exists because $R_{k+\ell}(\mu) \leq R_k(\mu) R_\ell(\mu)$, and $0 \leq \rho_\mu \leq M_\mu$.

Theorem 8 (Weighted-space part of Question 5). *If D is bounded on $L_\mu^\infty(T)$, then*

$$\sigma(C_b) = \{1\} \cup \{\lambda : |\lambda| \leq \rho_\mu\}, \quad \sigma(D) = \{0\} \cup \{\lambda : |\lambda - 1| \leq \rho_\mu\}.$$

Proof. Conjugate $L_\mu^\infty(T)$ isometrically to $\ell_\infty(T)$ by $(Jf)(v) = \mu(v)f(v)$, and group the coordinates by levels. Put $E_n = \ell_\infty(\{v : |v| = n\})$. The map from level $n - 1$ to level n is

$$(B_n x)(v) = \frac{\mu(v)}{\mu(v^-)} x(v^-).$$

On the nonroot tail $Z = (\bigoplus_{n \geq 1} E_n)_{\ell_\infty}$, the restriction of $JC_b J^{-1}$ has the form in Lemma 7. Moreover,

$$\|B_{n+k} \cdots B_{n+1}\| = \max_{\substack{|u|=n, v \succeq u \\ |v|-|u|=k}} \frac{\mu(v)}{\mu(u)},$$

so its spectral radius is exactly ρ_μ .

Splitting off the root coordinate gives the lower triangular block form

$$JC_b J^{-1} = \begin{pmatrix} 1 & 0 \\ B_1 & S \end{pmatrix} \quad \text{on } \mathbb{C} \oplus Z.$$

For $\lambda \neq 1$, elementary block elimination shows that $JC_b J^{-1} - \lambda I$ is invertible if and only if $S - \lambda I$ is invertible: the scalar block $1 - \lambda$ can be eliminated, leaving the diagonal blocks $1 - \lambda$ and $S - \lambda I$. For $\lambda = 1$, the first coordinate of every range vector of $JC_b J^{-1} - I$ is zero, so the operator is not onto. Consequently $\sigma(C_b) = \{1\} \cup \sigma(S)$, and Lemma 7 gives the first formula. Finally, $D = I - C_b$ and elementary spectral mapping give $\sigma(D) = 1 - \sigma(C_b)$, which is the second formula. $\square$

The singleton in each formula matters when $\rho_\mu < 1$: it records the root block and need not be absorbed by the disk.

9.2 Discrete Hardy spaces

Let T be $(q + 1)$ -homogeneous and $1 \leq p < \infty$. Write

$$c_0 = 1, \quad c_n = (q + 1)q^{n-1} \quad (n \geq 1),$$

and let E_n be the normalized finite-dimensional level space with norm

$$\|x\|_{E_n} = \left(\frac{1}{c_n} \sum_{|v|=n} |x(v)|^p \right)^{1/p}.$$

Then $\mathbb{T}_p(T)$ is isometrically $(\bigoplus_{n \geq 0} E_n)_{\ell_\infty}$.

Theorem 9 (Hardy-space part of Question 5). *For every $1 \leq p < \infty$,*

$$\sigma(C_b : \mathbb{T}_p \rightarrow \mathbb{T}_p) = \{\lambda : |\lambda| \leq 1\}, \quad \sigma(D : \mathbb{T}_p \rightarrow \mathbb{T}_p) = \{\lambda : |\lambda - 1| \leq 1\}.$$

Proof. The level map $B_n : E_{n-1} \rightarrow E_n$ replicates each parent value across its children. For $n \geq 2$, every level- $(n-1)$ vertex has q children and $c_n = qc_{n-1}$, so

$$\|B_n x\|_{E_n}^p = \frac{q}{c_n} \sum_{|u|=n-1} |x(u)|^p = \|x\|_{E_{n-1}}^p.$$

Thus every tail map and every consecutive tail product is an isometry. In Lemma 7, $R_k = 1$ and $\rho = 1$. Splitting off the root and applying the same scalar-block elimination used in Theorem 8 adds only the spectral point 1, which is already contained in the closed unit disk. Hence $\sigma(C_b) = \{\lambda : |\lambda| \leq 1\}$. Translating the spectrum by $D = I - C_b$ yields the disk centered at 1 of radius 1. $\square$

Theorems 8 and 9 completely answer Question 5. Together with Theorems 1, 3, 4, and 6, every question in the source list is settled.

10 Verification notes

The proof was subjected to the following checks.

1. The weighted space contains every point mass and all weights are positive, so the functional-space axiom is genuinely satisfied.
2. The exact U -isometry calculation includes the root column; the root copy is precisely why $\|C_b\| = \sqrt{2}$ rather than 1.
3. The lower bound for $\|D\|$ uses orthogonal level supports, not an unsupported approximate-eigenvector assertion.
4. In Lemma 2, the child one generation below the tested edge is essential and is supplied by the source's leafless/tree-child convention.
5. The compactness proof separately excludes finite-dimensional spaces before invoking the compact spectral theorem.
6. In Lemma 7, a one-level input forces all earlier solution coordinates to vanish, so the resolvent product estimate has no hidden homogeneous term.
7. The weighted radius starts at nonroot vertices because the root is represented separately by the scalar block 1.
8. The Hardy replication maps are exactly isometric after normalized level counting, including all consecutive products used by the shift lemma.

A reusable finite binary-tree check is included as `code/verify_weighted_hilbert.py`. At depth 10 it returns

$$\|C_b\| = 1.414213562373, \quad \|D\|_{\text{trunc}} = 1.979642883762, \quad \|Dy_{10}\|^2 = 3.7 = 4 - 3/10.$$

This finite computation is a consistency check only; the proofs above are exact and infinite-dimensional.

11 Scope, limitations, and novelty check

The packet fully answers all six questions under the definitions and tree conventions of the source. For the weighted-space spectrum, the formula is stated precisely in the bounded case, since an unbounded operator has no Banach-operator spectrum. The packet does not claim that a particular later paper has missed these observations; it records a candidate new solution pending expert priority review.

The bounded novelty search on 22 July 2026 covered:

- the run’s registry, solution, attempt, and proof-gap indexes;
- exact searches for arXiv:2207.10211, its title, and the source’s open question wording;
- close searches combining discrete differentiation on trees with power-boundedness, isometry, surjectivity, and compactness;
- exact and close searches for the weighted Banach and Hardy spectra, including operator-weighted unilateral shift and ℓ_∞ shift formulations;
- the later citing paper arXiv:2308.11016 on forward and backward shifts on Hardy spaces of a tree.

No later source explicitly stating any of these six resolutions was found. Classical operator-weighted shift literature is relevant background, but the shift lemma used here is proved directly and the exact application to Question 5 was not located. This supports moderate novelty confidence, not an exhaustive claim of priority.

Human review recommendation

Verify and circulate after a short priority search. The decisive checks are the exact norm calculation in Theorem 1, the pointwise-growth induction in Lemma 2, and the finite-dimensional step in Lemma 5. For the new spectral completion, check the resolvent estimate in Lemma 7 and the indexing of ρ_μ in Theorem 8.

References

- [1] R. F. Allen and C. M. Jackson, *The differentiation operator on discrete function spaces of a tree*, Involve **15** (2022), no. 1, 163–184; arXiv:2207.10211.
- [2] J. B. Conway, *A Course in Functional Analysis*, second edition, Graduate Texts in Mathematics 96, Springer-Verlag, New York, 1990.